HW5 readme

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RESULTS (es_test.conllu, 4128 transitions)

Micro F1: 0.9353
Macro F1: 0.6552
Weighted F1: 0.9319

APPROACH

Part 1 - Oracle

I adapted my HW4 arc-standard oracle to read CoNLL-U (using ID, FORM, LEMMA, UPOS, FEATS, HEAD, DEPREL and the comment lines are skipped). At each configuration it chooses:

LEFTARC(r) if s2 is not ROOT and the gold head of s2 is s1
RIGHTARC(r) if the gold head of s1 is s2 and s1 has no unattached dependents left
SHIFT otherwise

The RIGHTARC condition keeps a word on the stack until all of its dependents are attached. All the 843 training sentences were derived (0 skipped), giving 39,064 training instances. The transitions are written as SHIFT or (LEFTARC, label) / (RIGHTARC, label). The transitions are written as one transition per line with a blank line after each sentence. The root arc label comes out as "root".

Part 2 - Feature template

s = stack (s1 = top), b = buffer (b1 = front). A missing position gets the value NONE and the ROOT is given the form/lemma/UPOS "ROOT".

- form (lowercased), lemma, UPOS of s1, s2, s3
- UPOS of b1, b2, b3; form and lemma of b1
- FEATS (UD morphology string) of s1, s2, b1
- conjunctions: s1.upos+s2.upos, s1.form+s2.form, s1.upos+b1.upos, s2.upos+s1.upos+b1.upos, s1.form+b1.upos
- arc context: deprel of the leftmost and rightmost already-attached dependent of s1 and s2, and the number of attached dependents of each
- structure: stack depth (capped at 5), whether the buffer is empty, linear distance between s1 and s2 (4+ bucketed)

Rationale:

- The arc-standard acts on the top two stack items, so s1/s2 and their conjunctions carry most of the signal (e.g. DET above NOUN suggests LEFTARC det). A linear model cannot learn these interactions from separate features, hence the conjunctions.
- The buffer lookahead helps with SHIFT vs RIGHTARC: s1 should wait if the next word will attach to it (e.g. a following preposition or modifier).
- The RIGHTARC is only legal once s1 has all its dependents, so the attached-dependent labels and counts tell the model if s1 is "finished".
- The spanish nouns, adjectives and determiners agree in gender and number, so FEATS can help with attachments such as det, amod and nsubj.

- The lemmas generalize across inflected forms and the forms capture frequent function words.

Part 3 - Training and testing

I used the multinomial logistic regression from sklearn, with DictVectorizer to turn the feature dictionaries into sparse vectors with a maximum of 200 iterations. I first tried NLTK's MaxentClassifier (IIS), but it was very slow on about 39k instances with this many features, so I switched to sklearn for speed. At test time I step through each sentence with the GOLD transitions from es_test.seq.gold, predict a transition at every configuration with the same feature extractor, and write the predictions to hw5_pred.seq. The scores come from sklearn.metrics.f1_score on the flattened gold and predicted transition lists.

DISCUSSION

- The micro F1 (0.935) is much higher than the macro F1 (0.655). The micro pools all decisions, so frequent transitions like SHIFT and common arcs dominate it. The macro weights every label equally, so the many rarer dependency labels pull it down. I did not run a per-label error analysis, so I can't really say which labels are the worst.
- The micro-F1 equals accuracy, because each wrong prediction is both one false positive and one false negative with exactly one label per instance.
- One wrong prediction does not change later configurations since testing follows the gold path. These scores are therefore higher than a real parser's would be, since a real parser would also suffer from errors compounding.

ABLATIONS

I added an environment variable, HW5_ABLATE, that drops one group of features, and reran the full pipeline (training and testing) once per group.

Config	Micro F1	Macro F1	Weighted F1
Full template	0.9353	0.6552	0.9319
Without FEATS (morphology)	0.9339	0.6319	0.9302
Without lemmas	0.9348	0.6503	0.9313
Without forms	0.9334	0.6532	0.9299
Without conjunction features	0.9312	0.6466	0.9273
Without attached-dependent features	0.8929	0.5554	0.8855
Without lookahead (s3, b2, b3)	0.9348	0.6318	0.9310

Without structure features	0.9358	0.6454	0.9322

Group definitions: "forms" removes the form of s1/s2/s3/b1 and the two conjunctions that use s1.form; "conjunction features" removes all five conjunctions, so those two overlap with "forms". "attached-dependent features" are the leftmost/rightmost dependent labels and dependent counts of s1 and s2. "structure" is stack depth, buffer-empty and the s1-s2 distance.

Findings:

- The attached-dependent features matter the most as if you remove them, it drops micro F1 by 4.2 points (about 175 of 4128 transitions) and macro F1 by 10 points. Every other group changes micro F1 by less than half a point (at most about 17 transitions). My explanation is that the oracle only allows RIGHTARC once s1 has all of its dependents, so the model needs to know which dependents s1 already has to choose between SHIFT and RIGHTARC. I did not test this directly with an error analysis.
- The conjunction features were the next most useful (-0.4 micro), then forms (-0.2), FEATS (-0.1), and lemmas and lookahead (-0.05 each). I noticed that dropping the FEATS or the lookahead lowered macro F1 more (by about 2.3 points each) than micro F1, which may mean that they help mainly with the rarer labels, but I did not check this.
- One other thing that I noticed was that removing the structure features did not hurt the micro F1 (0.9358 vs 0.9353) while lowering macro F1 slightly, so they appear to add little. This is a difference of about 2 transitions.
- Caveat: this is a single test set of 4128 transitions with one run per setting, and I did not run significance tests, so the differences under about half a point (about 20 transitions) could be noise, so only the attached-dependent result holds value.

PROBLEMS AND WHAT I LEARNED

- The NLTK's MaxEnt training was too slow with this many features, which pushed me to learn about and use the sklearn's equivalent model.
- I learned how much the arc-standard RIGHTARC precondition matters, and how much difference removing features can make.